

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking data

CIM code	BS-PHYS-3
Program name	B.S. in Physics
Assessment Year	2024/2025
Program assessment coordinator	Todd Springer
Year of Current Assessment Plan	AY24
No assessment this year per plan	☐
LO(s) assessed	LO#3 and LO#4
Presented at program authority meeting date: <i>Please attach minutes</i>	Report was distributed to departmental leadership, assessment committee, and relevant instructors prior to submission

Methodology

According to the program assessment plan, in academic-year 2024/2025, the following learning objectives were assessed:

3. Demonstrate understanding of **quantum mechanics** by being able to solve problems (at an advanced undergraduate level) including problems involving
 - a. Bound states and/or scattering in simple one-dimensional potentials
 - b. Atomic physics (e.g. the hydrogen atom)
 - c. The quantum theory of angular momentum and spin

Relevant Courses: PHYS-223, PHYS-405, PHYS-406

4. Demonstrate understanding of **thermal and statistical physics** by being able to solve problems (at an advanced undergraduate level) including problems involving
 - a. The laws of thermodynamics
 - b. The connection between microscopic physics and thermodynamic properties
 - c. Boltzmann, Maxwell-Boltzmann, Fermi-Dirac, and Bose-Einstein distributions

Relevant Courses: PHYS-223, PHYS-304

Methodology for Assessment of Learning Objective #3

For the purposes of assessment of LO#3, artifacts were collected from PHYS-405: Fundamentals of Quantum Theory I. The artifacts were student submissions of the in-class final exam.

This course contained 14 students from several different programs. Each student on the roster was assigned an identifier (i.e. student 1 through student 14), but only students who are enrolled in this program are listed in the results sections below, which is why the numbers are not sequential. Also note that the rosters for PHYS-304 and PHYS-405 are not identical, so student #1 in the data table for LO#3 is not necessarily the same as student #1 in the data table for LO#4.

The problem statement is below

(10 pts) A quantum particle of mass M experiences a harmonic oscillator potential $V(x) = \frac{1}{2}M\omega^2x^2$. At time $t = 0$, the particle is in an equal superposition of ground state and first excited state. In other words, at time $t = 0$ if you measured the energy of the particle, you would have a $(2/3)$ probability of getting the ground state energy and a $(1/3)$ probability of getting the first excited state energy.

- (a) **(4 pts)** Write out the normalized initial wave function $\Psi(\xi, 0)$ using the dimensionless variable ξ instead of x . Your expression should be in terms of ξ , M , ω , and fundamental constants. If you use any other symbols, make sure they are defined clearly. It is ok to assume $\Psi(\xi, 0)$ is real and positive for all ξ .
- (b) **(4 pts)** What is the wave function at a later time $\Psi(\xi, t)$? Your expression in terms of ξ , M , ω , t and fundamental constants. If you use any other symbols, make sure they are defined clearly.
- (c) **(2 pts)** Does the probability density for this particle depend on time, or not? You can either explain your answer in words, or do a short calculation to justify your answer.

This is a classic problem in quantum mechanics, involving the harmonic oscillator potential. The problem asks students to demonstrate an understanding of wave functions, superposition, measurement probability, and Schrodinger time evolution. These central concepts are fundamental for the understanding of the postulates of quantum mechanics. Specifically, they are closely related to item 3a in the Learning Objectives (the harmonic oscillator is a simple one-dimensional potential).

The rubric used to assess these problems is given below

Problem or part	Weight	DOES NOT MEET EXPECTATIONS 1 - 2	MEETS WITH CONCERNS 3 - 4	MEETS EXPECTATIONS 5 - 7	EXCEEDS EXPECTATIONS 8
part a	0.4	weak attempt at a solution but with some relevant elements	major errors but includes some conceptual understanding e.g. of superposition	recognizes major concepts: superposition, ψ_0 and ψ_1 , coefficients, but makes minor mistakes	no mistakes; correct superposition with correct coefficients
part b	0.4	weak attempt at a solution but with some relevant elements	recognizes that there is an exponential evolution term but does not implement it correctly	adds exponential factor for each term but with minor mistakes	no mistakes; correct exponential factor for each term
part c	0.2	does not attempt an answer or answers no without or with an explanation that does not show understanding of how time dependence is determined	correctly answers that it depends on time, but does not explain or says no but offers an explanation that shows some understanding of what determines time dependence	correctly answers that it depends on time and makes some vague attempt at explaining	correctly answers that it depends on time and properly explains or shows why

Methodology for Assessment of Learning Objective #4

For the purposes of assessment of LO#4, artifacts were collected from PHYS-304 – Thermodynamics and Statistical Physics. The artifacts were student final exams.

This course contained 13 students. Each student on the roster was assigned an identifier (i.e. student 1 through student 13), but only students who are enrolled in this program are listed in the results sections below, which is why the numbers are not sequential.

The following two problems were assessed:

Problem 1 (15 points): An insulated cylinder contains 5.0 liters of air at room temperature (300K) and atmospheric pressure (10^5 N/m^2). At one end of the cylinder is a massless piston, whose surface area is 0.01 m^2 . Suppose that you push the piston in very hard, exerting a force of 2500N. The piston moves 4mm before it is stopped.

- Calculate the amount of work you did on the system, and the heat added to the gas. Assuming that all of the energy added goes into the gas (not the piston or cylinder walls), by how much does the internal energy of the gas increase?
- Calculate the change in entropy of the gas once it has again reached equilibrium. (Because the piston displacement is tiny compared with the total volume of the gas, you may assume the internal pressure and temperature of the gas are constant in this problem.
- Is this process reversible or irreversible? Explain. If it is reversible, how could you make it irreversible? Likewise, if it is irreversible, how could you make it reversible?

Problem 2 (15 points): The multiplicity for a cold, large Einstein solid is:

$$\Omega = \left(\frac{e N}{q} \right)^q$$

With the system's total energy defined by $U = \epsilon q$.

- Calculate the temperature of the solid as a function of its total energy
- Calculate the energy of the solid as a function of temperature.
- Calculate the constant volume heat capacity of the solid.

Problem #1 assesses whether students understand thermodynamics processes and the first and second laws of thermodynamics. Part (a) deals primarily with the first law, while parts (b) and (c) also involve the second law of thermodynamics. Problem #2 involves the connection between microscopic physics (multiplicity and microstates) and thermodynamic observables like temperature and heat capacity. As such both problems are well-aligned with LO#4 which explicitly mentions the laws of thermodynamics and the micro/macro connection. Deep understanding of these topics and mastery of problems on these topics are central to our program's learning goals.

The rubric used to assess both problems is given below

Problem or part	Weight	DOES NOT MEET EXPECTATIONS 1 - 3	MEETS WITH CONCERNS 4 - 5	MEETS EXPECTATIONS 6- 7	EXCEEDS EXPECTATIONS 8
1a	0.2	Solution does not demonstrate an understanding of the first law of thermodynamics, or possesses multiple serious conceptual issues about the 1st law.	Some significant conceptual misunderstandings about internal energy, heat, and work, and their relationship.	Demonstrates significant understanding of most, if not all of internal energy, work, heat. Is able to relate these quantities with the 1st law	Demonstrates excellent understanding of internal energy, heat, and work, and how the 1st law of thermodynamics relates them
1b	0.2	Solution does not demonstrate knowledge or ability to solve problems involving entropy change	Some significant conceptual misunderstandings about entropy and how it can be related to other state variables	Demonstrates significant understanding of how the combined 1st and 2nd laws can be used to calculate entropy change, with minor calculational errors.	Demonstrates excellent understanding of how the combined 1st and 2nd laws can be used to calculate entropy change
1c	0.2	Demonstrates little understanding of the second law, entropy change, or reversibility	Demonstrates some understanding of how reversibility is related to entropy change and quasistatic processes, with significant conceptual errors	Demonstrates significant understanding of how reversibility is related to entropy change and quasistatic processes, with minor errors	Demonstrates excellent understanding of how reversibility is related to entropy change and quasistatic processes
2 (setup)	0.2	Student has great difficulty relating multiplicity, entropy, and heat capacity	Student has some difficulty relating multiplicity, entropy, and heat capacity with formulas; significant errors or omissions are present	Student can relate multiplicity, entropy, and heat capacity with formulas with only minor errors	Student can relate multiplicity, entropy, and heat capacity with formulas which are exactly correct
2 (math)	0.2	Student lacks sufficient mathematical skill to effectively solve the problems	Student demonstrates some knowledge of how to apply mathematical formulas correctly, but has significant difficulties in solving the problems	Student demonstrates significant calculus and algebraic skill, with minor errors	Student handles mathematical manipulations (partial derivatives and algebraic manipulations) easily and fully simplifies the results

Each artifact was evaluated according to the above rubric(s), and a weighted average was used to arrive at a final score ranging from 0 to 8. A letter grade was assigned for each Learning Objective as follows:

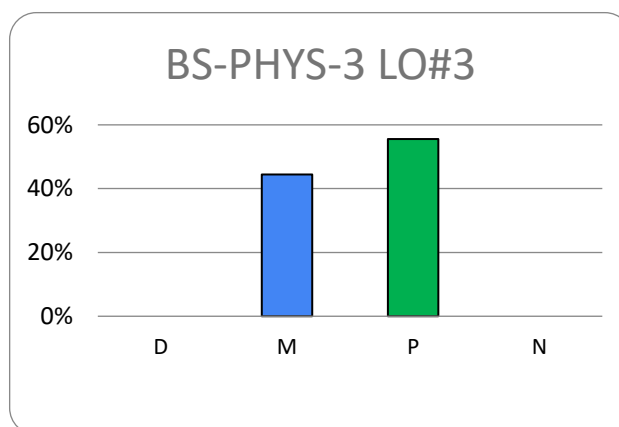
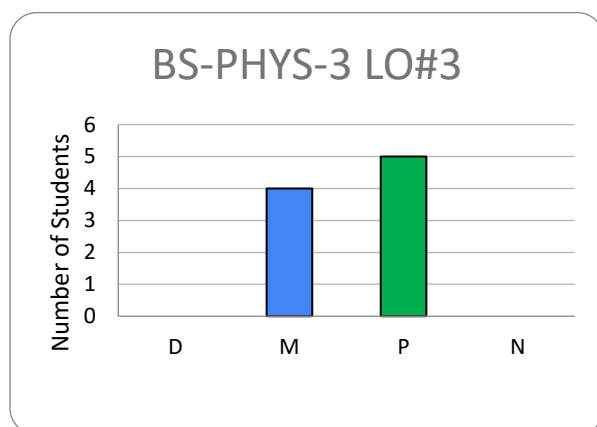
Designation	Score Range
D (Does Not Meet Expectations)	0 - 4
M (Meets Expectations)	4 - 6.5
P (Proficient)	6.5 – 8.0

*Note that a student receives N (Not Assessed) in cases where the student withdrew from the course, took an incomplete, or when an artifact is not available due to issues outside of the student's control. A student who is non-compliant (i.e. does not submit an assigned assessment) receives D (Does not Meet Expectations).

Results for LO#3 – Quantum Mechanics

The classlists for PHYS-304 and PHYS-405 were not identical; student #1 in the LO#3 data table is not necessarily the same as student #1 in the data table for LO#4

student	major	D/M/P
student 1	ASPY/PHYS	P
student 4	PHYS	M
student 6	PHYS	M
student 7	PHYS	M
student 8	PHYS	P
student 9	PHYS	P
student 10	PHYS	P
student 12	PHYS	P
student 14	PHYS	M



Problem Part	Average Percentage Score
a	82%
b	94%
c	67%

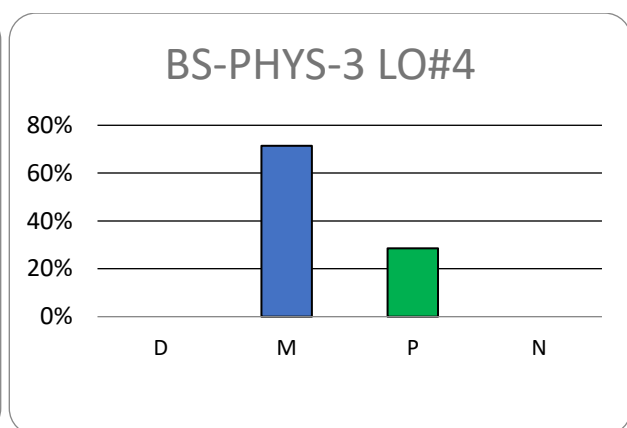
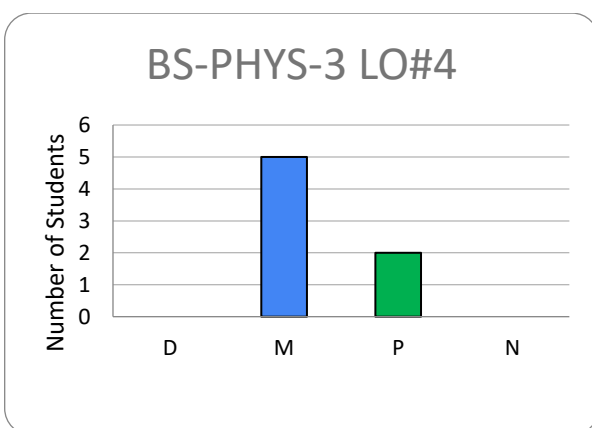
Artifact samples for LO#3 – Quantum Mechanics

Two example artifacts are included with this report

- Student 8 - Assessed as Proficient (P)
- Student 6 - Assessed as Meets Expectations (M)
- No students were assessed as Does Not Meet Expectations (D), so no artifact of this classification can be provided

Results for LO#4 – Thermal and Statistical Physics

student	major	D/M/P
student 1	PHYS	P
student 4	PHYS	M
student 5	ASPY/PHYS	M
student 6	PHYS	M
student 7	PHYS	M
student 9	PHYS	M
student 10	PHYS	P



The overall averages by part (scored according to the rubric) are given below

Problem Part	Average Percentage Score
1a	70%
1b	66%
1c	73%
2 (setup)	86%
2 (math)	59%

Artifact samples for LO#4 – Thermal and Statistical Physics

Two example artifacts are included with this report

- Student 10 - Assessed as Proficient (P)
- Student 6 - Assessed as Meets Expectations (M)
- No students were assessed as Does Not Meet Expectations (D)

Description of findings

LO#3

As all assessed students were graded as either (M) or (P), there is no evidence in the data that students are not meeting the desired learning outcome.

When the data is analyzed in finer detail (broken down by problem part), students scored significantly lower on part (c) (an average of 67%) than on the other scored parts. This part is different than the other parts in that it allows students to express their answer in words rather than by doing a calculation. The lower scores may be an indication that students are comfortable with calculations but may lack deeper conceptual understanding that can be expressed outside of a calculation. It may also indicate that students are not practiced at interpreting formulas or equations in everyday language.

LO#4

As all assessed students were graded as either (M) or (P), there is no evidence in the data that students are not meeting the desired learning outcome.

When the data is analyzed in finer detail (broken down by problem part), students scored significantly lower on the mathematical manipulations of problem #2 (an average of 59%) than on the other scored parts. There is some question as to whether these math skills should be assessed as part of this learning objective, or if they fall under LO#5 (to be assessed in AY26). There is an argument to be made that LO#4 explicitly mentions problem solving, and problem solving necessarily involves some mathematical manipulations.

Continuous Improvement Plans

Prior CI plan impact

☒ no previous CI plans available

This cycle's CI plan

LO#3: To address the significantly lower scores on part (c), we recommend students gain additional practice interpreting formulas, describing the results of calculations in words, and reflecting on their mathematical results. Additional reflection questions could be asked as part of regular homework assignments. The instructor has agreed to add such “reflection questions” on homework assignments the next time the course is taught.

We can examine whether this CI plan is successful in the next assessment cycle for this learning objective. Ideally, the same final exam problem would be given and scored according to the same rubric. The CI plan would be considered successful if the average student score on part (c) is in-line with the averages on the other assessed parts.

LO#4: The mathematical skills needed for Problem #2 should be covered in prerequisite precalculus and calculus courses. Nevertheless, the use of logarithms and exponentials may show up more frequently in statistical mechanics than in other courses, so students may need additional support in the form of review in lecture, explicit homework problems about rules for logarithms and exponentials, and increased emphasis on these mathematical steps. The instructor has stated that they will attempt to increase this emphasis next time the course is taught.

We can examine whether this CI plan is successful in the next assessment cycle for this learning objective. Ideally, the same final exam problem would be given and scored according to the same rubric. The CI plan would be considered successful if the average student math score on problem #2 is in-line with the averages on the other assessed parts.